

Learning Dynamics

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Stormglass

Big questions

- How do models generalize?
- How does internal structure form during training?
- Emergent capabilities
- Control levers for the training process
- How do the loss, architecture, data, and optimizer determine which internal structures are learned throughout training?

How: making mathematical models of learning

- Tractable toy models: deep linear networks, shallow nonlinear networks, polynomial activations
- Take limits in depth, width, data, and training time
- Exploit symmetries: quotient them out and find conserved quantities
- Identify useful coarse-grained variables: singular values of weights, learning coefficients, correlation functions
- Work in specific regimes: lazy versus rich
- Validate with tight feedback loops: experiments, symbolic derivations, and mathematical proofs

Safety motivations

- A central question in AI safety is whether the training process selects a harmful model, such as a scheming, sycophantic, or spiteful model
- Predict whether a model will be harmful in high-stakes, out-of-distribution environments when behavioural or average-case interpretability evaluations cannot be trusted
- Develop stronger interpretability methods by decomposing models throughout training
- We want a mathematical model that can outperform average-case methods at predicting harmful behaviour
- More ambitiously, we would like to find control levers in the training process that make us confident training will select a safe model

Setup

Deep neural network

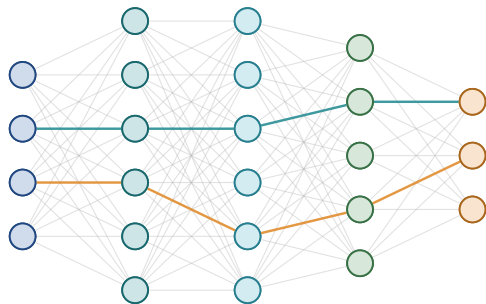
Let $\theta \in \Omega$ be the parameters, let $x \in \mathbb{R}^{d_0}$, and let $W_\ell \in \mathbb{R}^{d_\ell \times d_{\ell-1}}$ with biases b_ℓ and activation σ . With $h_0 = x$,

$$h_\ell(\theta, x) = \sigma(W_\ell h_{\ell-1}(\theta, x) + b_\ell), \\ \ell = 1, \dots, L-1.$$

A neural network is a function f_θ such that

$$f_\theta(x) = W_L h_{L-1}(\theta, x) + b_L.$$

Other architectural components include residual streams, tokenization, embeddings, attention for sequence modeling, and layer normalization.



Deep linear network

The function is linear in the data and nonlinear in the parameters:

$$f: \mathbb{R}^{d_0} \times \prod_{\ell=1}^L \mathbb{R}^{d_\ell \times d_{\ell-1}} \longrightarrow \mathbb{R}^{d_L},$$
$$(x, W_1, \dots, W_L) \longmapsto W_L \cdots W_1 x = Wx.$$

d_0 is the input dimension; d_ℓ , $1 \leq \ell \leq L$, is the width of layer ℓ . Define the product map:

$$\mu(\theta) := W_L \cdots W_1 = W,$$

where W is the student matrix.

Stochastic gradient descent

Learn a function g from observations $y = g(x)$. Initialize neural weights at $\theta_0 \sim \mathcal{N}(0, \sigma^{-\gamma} I)$, with initialization scale γ .

SGD update

Consider a dataset of size N . At each time step, sample a batch b_k of size B . The empirical loss is the expectation of the batch loss:

$$\mathcal{L}(\theta) = \mathbb{E}_b[\mathcal{L}(\theta; b)].$$

The SGD update is

$$\begin{aligned}\theta_{k+1} &= \theta_k - \eta \nabla_{\theta} \mathcal{L}(\theta_k; b_k) \\ &= \theta_k - \eta \nabla_{\theta} \mathcal{L}(\theta_k) + \eta \xi(\theta_k, b_k).\end{aligned}$$

with gradient noise

$$\xi(\theta, b) := \nabla_{\theta} \mathcal{L}(\theta) - \nabla_{\theta} \mathcal{L}(\theta; b).$$

The loss $\mathcal{L}$ can be, for example, mean squared error or cross-entropy loss.

Data, loss function, gradient flow

Quadratic population loss and gradient flow

Let M be the teacher matrix, with $y = Mx$ and $x \sim \mathcal{N}(0, I_{d_0})$. The input-output covariance is $\Sigma_{YX} = \mathbb{E}[yx^\top] = M$.

$$\mathcal{L}(\theta) := \frac{1}{2} \|M - W_L \cdots W_1\|_F^2 = \frac{1}{2} \|M - W\|_F^2$$

Initialize $\theta_0 \sim \mathcal{N}(0, \sigma^{-\gamma} I)$, where γ controls the initialization scale. The gradient flow is

$$\dot{W}_\ell = -\nabla_{W_\ell} \mathcal{L}(\theta).$$

Frobenius norm: $\|A\|_F := \sqrt{\text{Tr}(A^\top A)}$.

Global minimizers achieve zero loss: $\exists \theta^* \in \Omega$, $\mu(\theta^*) = W^* = M$.

The structure of the data is encoded by Σ_{YX} .

Geometry

Global minima and saddle point

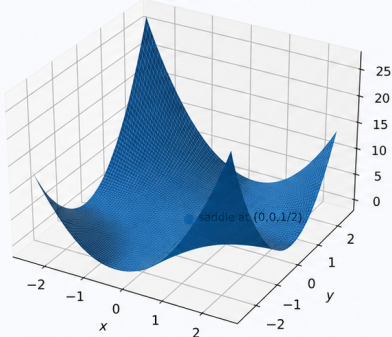
Loss: $\ell_1(x, y) = \frac{1}{2}(xy - 1)^2$

Global minima: $\{(x, y) : xy = 1\}, \ell_1 = 0$

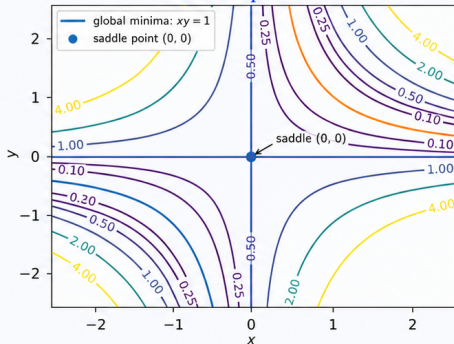
Saddle point: $(0, 0), \ell_1(0, 0) = \frac{1}{2}$

Strict saddle: along $x = y = t$ the loss decreases near 0, while along $x = -y = t$ the loss increases near 0

3D loss surface



Contour plot



Global minima form the hyperbola $xy = 1$, while $(0, 0)$ is a strict saddle point.

Deep learning without poor local minima

Assume that Σ_X and Σ_{YX} are full rank, and that Σ_{YX} has d_L distinct singular values.

No non-global local minima

The loss function of a DLN of depth L is non-convex and non-concave, and every critical point that is not a global minimum is a saddle point.

The training dynamics cannot get stuck in spurious local minima.

Reference: *Kawaguchi, Kenji. "Deep learning without poor local minima." Advances in neural information processing systems 29 (2016)*

Parametrization of critical points

Teacher SVD: $M = \sum_{i=1}^{r_m} s_i u_i v_i^\top$.

Saddle points

Let $\theta \in \Omega$ be a critical point of rank r . Then there exists a subset S of left singular vectors of M and a projector P_S onto their span such that

$$P_S := \sum_{i \in S} u_i u_i^\top, \quad \mu(\theta) = P_S M = \sum_{i \in S} s_i u_i v_i^\top.$$

The saddle points are given by a subset of singular values and (left) singular vectors of the teacher M .

Reference: Achour, El Mehdi, François Malgouyres, and Sébastien Gerchinovitz. "The loss landscape of deep linear neural networks: a second-order analysis." *Journal of Machine Learning Research* 25.242 (2024): 1-76.

Symmetries

Let $G_{\mathbf{d}} := \prod_{\ell=0}^L \mathrm{GL}_{d_{\ell}}$ act on Ω via

$$(W_1, \dots, W_L) \mapsto (g_1 W_1 g_0^{-1}, g_2 W_2 g_1^{-1}, \dots, g_L W_L g_{L-1}^{-1}).$$

Equivariance of the multiplication map

Since $W_L g_{L-1}^{-1} g_{L-1} W_{L-1} \cdots g_1^{-1} g_1 W_1 = W_L \cdots W_1$, the multiplication map μ is $G_{\mathbf{d}}$ -equivariant and invariant under the hidden gauge group G_h :

$$\mu(g \cdot \theta) = g_L \mu(\theta) g_0^{-1}, \quad \mu(g_h \cdot \theta) = \mu(\theta).$$

A DLN is degenerate: infinitely many parameter values can map to the same student matrix.

It is natural to study solutions up to $G_{\mathbf{d}}$ symmetries. By equivariance of μ , $G_{\mathbf{d}}$ acts on the determinantal variety

$$\Sigma_{\mathbf{d}}^r := \{\theta \in \Omega \mid \mathrm{rank}(\mu(\theta)) = r\}.$$

Stratification of the fiber of the product map

Orbits are parameterized by rank patterns

The $G_{\mathbf{d}}$ -orbits in $\Sigma_{\mathbf{d}}^r$ are indexed by rank patterns

$$\rho_{ij} := \text{rank}(W_j \cdots W_i), \quad 1 \leq i \leq j \leq L.$$

The corresponding orbit decomposition is

$$\Sigma_{\mathbf{d}}^r = \bigsqcup_{\rho} \mathcal{O}_{\rho}.$$

Two DLNs are in the same $G_{\mathbf{d}}$ -orbit if and only if they have the same rank pattern.

Since $\Sigma_{\mathbf{d}}^r = G_{\text{out}} \cdot \mu^{-1}(W)$, this gives a combinatorial description of the fiber of the product map $\mu^{-1}(W)$.

Rank patterns are natural observables arising from geometric invariants of DLNs.

Codimension of the global minima

For a teacher matrix M of rank r_m , we have:

$$\begin{aligned}\operatorname{codim}(\mu^{-1}(M)) &= \operatorname{codim}(\Sigma_{\mathbf{d}}^{r_m}) + r_m(d_0 - d_L - r_m) \\ &= 2 \operatorname{RLCT}_0(\mathcal{L}).\end{aligned}$$

In general, we only have:

$$\operatorname{RLCT}_0(\mathcal{L}) \leq \frac{1}{2} \operatorname{codim}_{\operatorname{Rep}_{\mathbf{d}}}(\mathcal{L}^{-1}(0)).$$

Ref: *Lehalleur, Simon Pepin, and Richárd Rimányi. "Geometry of fibers of the multiplication map of deep linear neural networks." arXiv preprint arXiv:2411.19920 (2024).*

Takeaways

Under full-rank and nondegenerate-data assumptions:

- Critical points of DLNs are either saddles or global minima
- Gauge symmetries mean there is a whole algebraic variety of degenerate minima
- Geometric invariants give observables and a combinatorial description of the parameter-function map and the global minima

- Kenji Kawaguchi. Deep learning without poor local minima. *Advances in Neural Information Processing Systems*, 29, 2016.
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Dynamics

Gradient flow and NTK dynamics

In the small-learning-rate limit, gradient descent becomes gradient flow:

$$\theta_{t+1} = \theta_t - \eta \nabla_{\theta} \mathcal{L}(\theta_t) \quad \rightsquigarrow \quad \dot{\theta}(t) = -\nabla_{\theta} \mathcal{L}(\theta(t))$$

Neural Tangent Kernel (NTK):

$$K_{\theta}(x, x') = \nabla_{\theta} f_{\theta}(x) \nabla_{\theta} f_{\theta}(x')^{\top} = \sum_{i=1}^p \partial_{\theta_i} f_{\theta}(x) \partial_{\theta_i} f_{\theta}(x')^{\top}.$$

For a quadratic loss on a dataset (x_i, y_i) , the dynamics in function space are

$$\frac{d}{dt} f_{\theta_t}(x) = - \sum_{i=1}^N K_{\theta_t}(x, x_i) (f_{\theta_t}(x_i) - y_i).$$

Conserved quantities in DLNs

Conserved quantities under gradient flow:

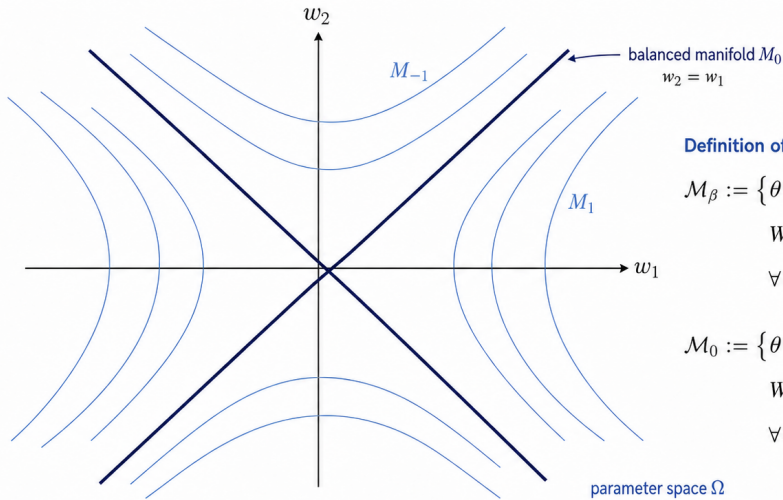
$$G_\ell := W_{\ell+1}^\top W_{\ell+1} - W_\ell W_\ell^\top, \quad \ell = 1, \dots, L-1.$$

Along the flow, $G_\ell = \beta_\ell$ is constant. When $\beta_\ell = 0$, we have **balanced weights**.

For a one-dimensional, two-layer DLN, we get a hyperbolic equation:

$$w_2^2 - w_1^2 = \beta.$$

Foliation of parameter space by balance leaves



Definition of the balance leaves

$$\mathcal{M}_\beta := \left\{ \theta \in \Omega : \begin{aligned} &W_{\ell+1}^\top W_{\ell+1} - W_\ell W_\ell^\top = \beta_\ell, \\ &\forall \ell = 1, \dots, L-1 \end{aligned} \right\}$$

$$\mathcal{M}_0 := \left\{ \theta \in \Omega : \begin{aligned} &W_{\ell+1}^\top W_{\ell+1} - W_\ell W_\ell^\top = 0, \\ &\forall \ell = 1, \dots, L-1 \end{aligned} \right\}$$

► Gradient flow remains on a fixed balance leaf.

The manifold of gradient flow

Gradient flow is constrained to the following hyperbolic manifold:

$$\mathcal{M}_\beta := \{\theta \in \Omega \mid W_{\ell+1}^\top W_{\ell+1} - W_\ell W_\ell^\top = \beta_\ell, \ell = 1, \dots, L-1\}.$$

Gradient flow on the balanced variety

On the balanced variety $\mathcal{M}_0$, the self-consistent gradient flow equation in function space is

$$\dot{W} = \sum_{k=1}^L (WW^\top)^{\frac{L-k}{L}} R(W) (W^\top W)^{\frac{k-1}{L}}, \quad R(W) := \Sigma_{YX} - W.$$

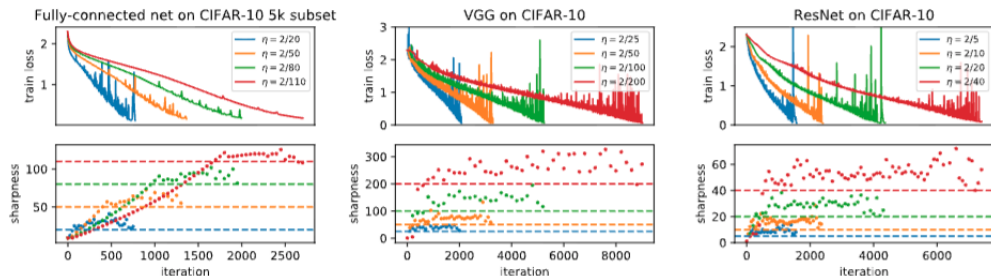
For depth $L = 2$:

$$\mathcal{M}_0 = \{(W_1, W_2) \mid W_2^\top W_2 = W_1 W_1^\top\}.$$

$$\dot{W} = (WW^\top)^{\frac{1}{2}} R(W) + R(W) (W^\top W)^{\frac{1}{2}}.$$

Learning rate: Edge of stability

Descent lemma: for an L_{smooth} -smooth convex loss, the loss decreases if $0 < \eta < 2/L_{\text{smooth}}$. The loss is nonconvex, but the Hessian oscillates near and slightly above $\frac{2}{\eta}$; this regularizes away sharp minima.



Ref: Gradient Descent on Neural Networks Typically Occurs at the Edge of Stability, Cohen et al.

Key points

- NTK preconditions gradient flow in function space
- Symmetries can induce conserved quantities that constrain the gradient flow
- Gradient flow in function space for balanced quantities leads to self-consistent equations
- The effect of the learning rate can be understood via the edge of stability

- Menon, Govind. The geometry of the deep linear network. Symposium on Probability and Stochastic Processes. Cham: Springer Nature Switzerland, 2023.
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- Arthur Jacot, Franck Gabriel, and Clément Hongler. Neural tangent kernel: Convergence and generalization in neural networks. *Advances in Neural Information Processing Systems*, 31, 2018.
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Initialization, Width, and Depth

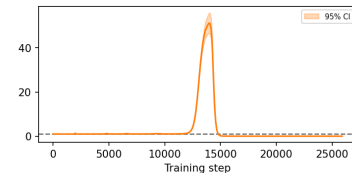
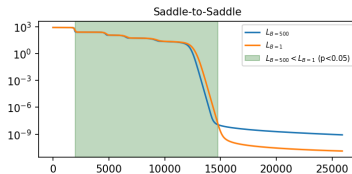
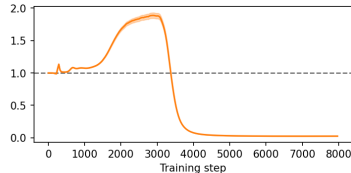
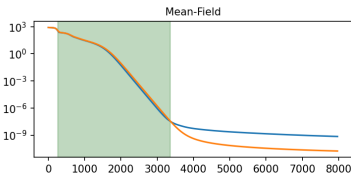
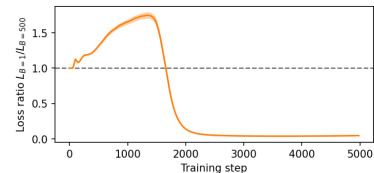
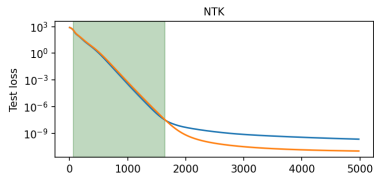
Training losses

Initialize weights: $\theta \sim \mathcal{N}(0, \sigma^{-\gamma/2} I)$

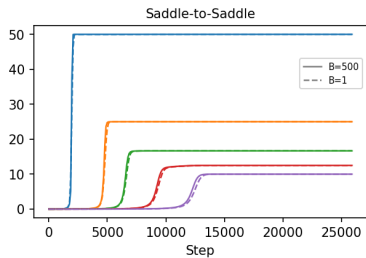
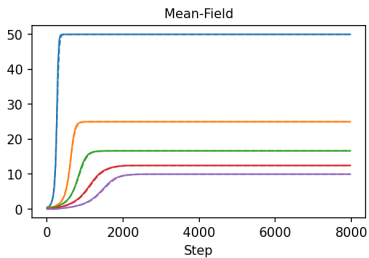
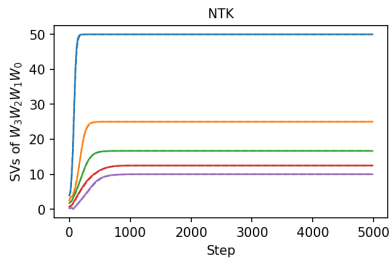
$\gamma < 1$: large initialization, linear regime (NTK)

$\gamma > 1$: small initialization, saddle-to-saddle

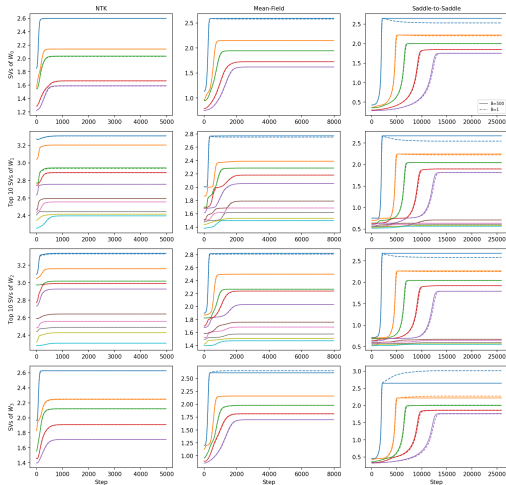
$\gamma = 1$: intermediate regime



Singular values of the student matrix



Singular values of individual weight matrices



Lazy regime (NTK or linear regime)

For a DLN of depth L with initialization $\theta(0) \sim \mathcal{N}(0, \sigma^2 I)$ and $\sigma^2 = d_L^{-\gamma}$, with $\gamma < 1$:

$$\dot{W} \approx K_0(M - W).$$

This leads to the following solution:

$$W(t) = M - \exp(-K_0 t)(M - W(0)).$$

The timescale of learning is fixed by the NTK at initialization:

$$t \approx -k_0^{-1} \log \left(\frac{s - w_f}{s - w_0} \right)$$

This behaves like a shallow linear network.

Rich regime

For balanced and aligned weights (or $\gamma > 1$), the singular values of the student satisfy

$$\dot{w}_\alpha = 2(s_\alpha - w_\alpha) w_\alpha.$$

Solving this ODE for $w(t) := w_\alpha$ gives:

$$w(t) = \frac{s}{1 + \left(\frac{s}{w_0} - 1\right) e^{-2st}}.$$

The timescale of learning is

$$t = \frac{1}{2s} \ln \left(\frac{w_f (s - w_0)}{w_0 (s - w_f)} \right), \quad s \neq 0.$$

Key points

- Depth and specific initialization induce saddle-to-saddle dynamics
- Gradient flow has a simplicity bias toward low-rank approximation
- Large initialization and large width give the NTK regime: no feature learning, and the NTK is frozen
- In the saddle-to-saddle regime, the timescale of learning is set by the target singular values

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- Jacot, Arthur, et al. Saddle-to-saddle dynamics in deep linear networks: Small initialization training, symmetry, and sparsity. *arXiv preprint arXiv:2106.15933* (2021).
- Dominé, Clémentine CJ, et al. From lazy to rich: Exact learning dynamics in deep linear networks. *arXiv preprint arXiv:2409.14623* (2024).
- Zhang, Yedi, Andrew Saxe, and Peter E. Latham. Saddle-to-Saddle Dynamics Explains A Simplicity Bias Across Neural Network Architectures. *arXiv preprint arXiv:2512.20607* (2025).
- Zhenfeng Tu, Santiago Tomas Aranguri Diaz, and Arthur Jacot. Mixed dynamics in linear networks: Unifying the lazy and active regimes. *Advances in Neural Information Processing Systems*, 37:106059–106104, 2024.

Stochasticity

- SGD: Sample a batch from a dataset and compute the batch gradient
- Continuous model with Brownian motion B_t

$$d\theta_t = -\nabla \mathcal{L}(\theta_t) dt + \sqrt{\eta} \Sigma^{1/2}(\theta_t) dB_t.$$

- The gradient-noise covariance $\Sigma(\theta_t)$ depends on geometry: it is anisotropic and state-dependent

Fokker-Planck: Boltzmann equilibria

The Fokker-Planck equation (for density $p(\theta, t)$) is a local conservation law:

$$\begin{aligned}\partial_t p(\theta, t) &= -\nabla \cdot j(\theta, t), \\ j(\theta, t) &= -p(\theta, t) \nabla \mathcal{L}(\theta) - \frac{\eta}{2} \nabla_{\theta}^{\top} (\Sigma(\theta) p(\theta, t)).\end{aligned}$$

Important special case:

- Stationary distribution: $\partial_t p^*(\theta) = 0$
- Thermal equilibrium: $j = 0$
- White noise: $\Sigma = \sigma^2 I$

$$p^*(\theta) \propto \exp\left(-\frac{2}{\eta\sigma^2} \mathcal{L}_N(\theta)\right) \quad (\text{Boltzmann}).$$

Noise anisotropy

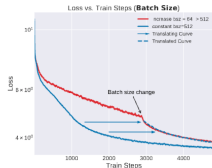
- In general, noise is anisotropic: it introduces a different implicit bias
- In function space, Langevin dynamics introduce an additional drift

$$\frac{\eta}{2} \text{Tr}(\Sigma(\theta)H(\theta))$$

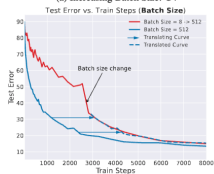
- Heuristic: bias toward flatter minima
- More:
 - See Govind Menon On the implicit regularization of Langevin dynamics with projected noise
 - Our paper deriving Langevin dynamics for DLNs during the saddle-to-saddle regime

Golden Path hypothesis

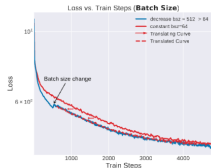
- Lots of debate about the implicit bias of SGD noise
- Transient regime (drift-dominated) versus low-loss regime (diffusion-dominated):



(a) Increasing Batch Size: C4



(c) Increasing Batch Size: CIFAR 5m



(b) Decreasing Batch Size: C4

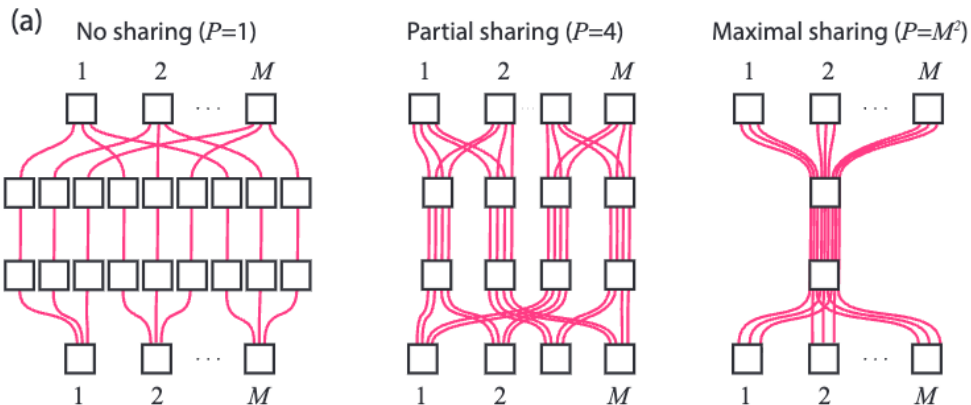


(d) Decreasing Batch Size: CIFAR 5m

Beyond Implicit Bias: The Insignificance of SGD Noise in Online Learning

Towards nonlinearity

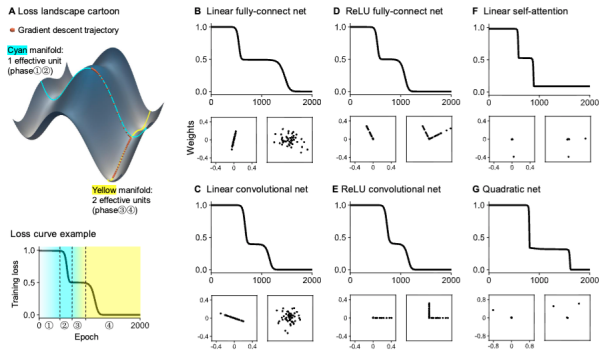
Neural race reduction



The Neural Race Reduction: Dynamics of Abstraction in Gated Networks, Andrew M. Saxe et al.

MLP and linear transformers

Saddle-to-Saddle Dynamics Explains A Simplicity Bias Across Neural Network Architectures, Saxe et al.



Key points

- SGD can be modeled by an SDE with state-dependent noise
- If the noise is white and the system is at thermal equilibrium, the stationary distribution of the SDE is Boltzmann
- The implicit bias of stochasticity is not fully understood (flatness?), but it is more of a second-order effect
- Nonlinearity can be modeled with gated DLNs: neural race reduction
- MLPs and linear transformers also have saddle-to-saddle dynamics

Promising directions

- Polynomial neural networks (quadratic activations)
- Related: tensor networks and tree tensor networks

Big questions

- Toy models of deep nonlinear learning
- Theory that captures natural data
- Making explicit the implicit biases of realistic architectures
- Interplay between geometry and learning
- A statistical field theory of deep neural networks
- A more realistic theory of feature and hierarchical learning

Reference: *There Will Be a Scientific Theory of Deep Learning*, Jamie Simon et al.